

1D CNN — Architecture Variant Comparison

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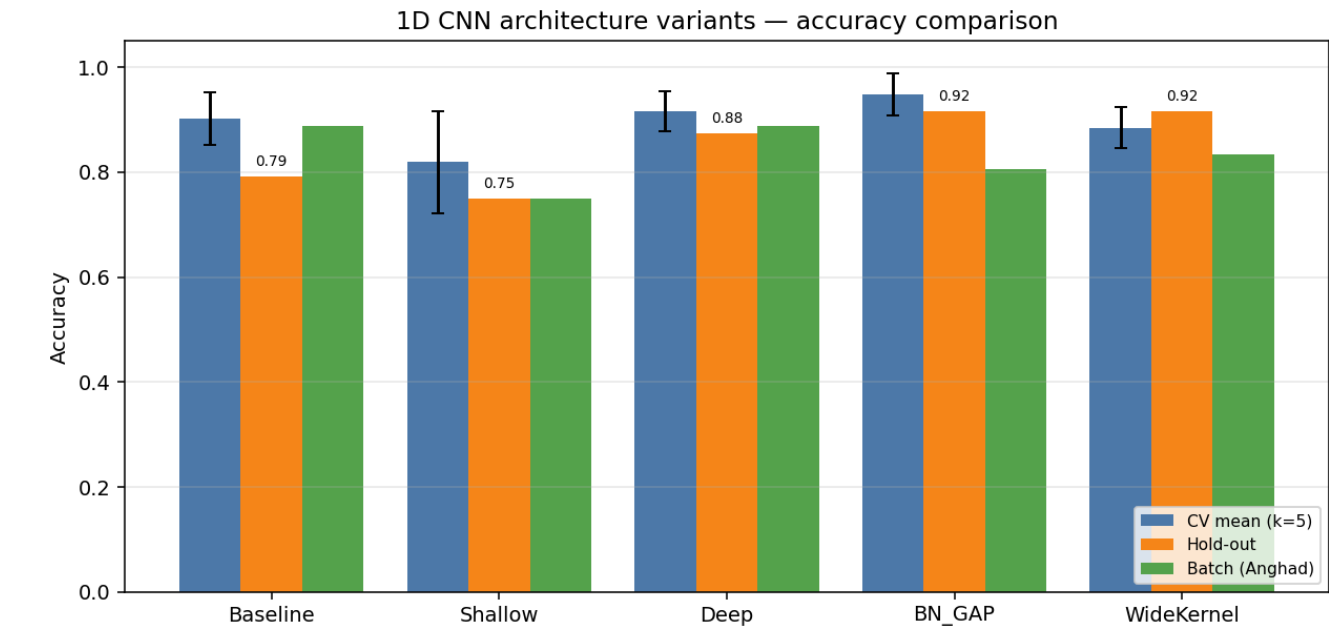
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/Harry/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/Bella/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/Raquel/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/Tash/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/Anghad/Dynamic
Total trials	240 (train pool 216, hold-out 24)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

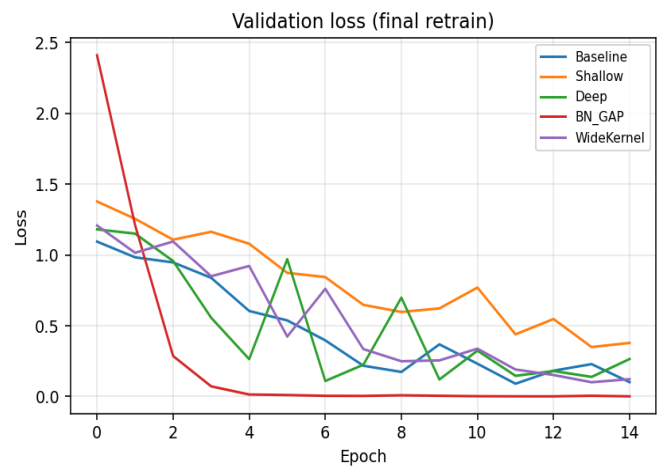
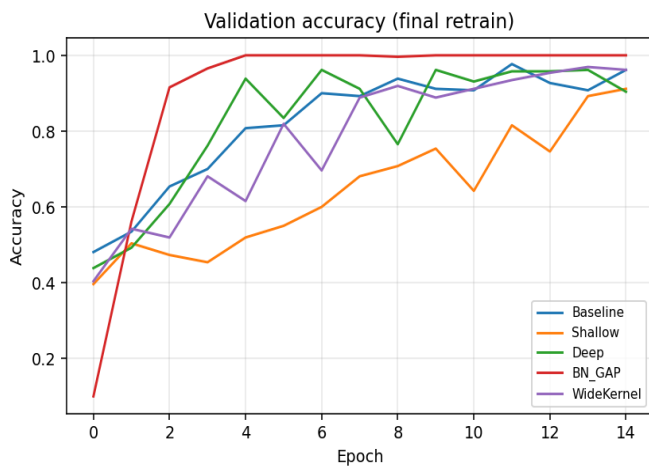
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9026	0.0498	0.7917	0.4309	0.8889	32/36
Shallow	72,487	0.8191	0.0973	0.7500	0.8049	0.7500	27/36
Deep	204,007	0.9165	0.0380	0.8750	0.6679	0.8889	32/36
BN_GAP	57,319	0.9490	0.0400	0.9167	0.1766	0.8056	29/36
WideKernel	131,687	0.8841	0.0392	0.9167	0.5044	0.8333	30/36

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9026 ± 0.0498
Hold-out test acc / loss	0.7917 · 0.4309
Batch-test acc	0.8889 (32/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9318	0.9318	0.9318
2	1038	43	0.9535	0.9535	0.9535
3	1038	43	0.8140	0.8605	0.8140
4	1038	43	0.9302	0.9302	0.9302
5	1038	43	0.8837	0.9302	0.8837

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	3
Double_Nothing	0.333	0.333	0.333	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.600	1.000	0.750	3
Double_cmere	1.000	0.750	0.857	4
Drum_Roll	1.000	0.333	0.500	3
macro avg	0.819	0.774	0.761	24
weighted avg	0.833	0.792	0.781	24

Batch test per class (Anghad)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	1	5	0.200
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	6	6	1.000
Drum_Roll	5	5	1.000
Overall	32	36	0.889

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8191 ± 0.0973
Hold-out test acc / loss	0.7500 · 0.8049
Batch-test acc	0.7500 (27/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.8864	0.8864	0.8864
2	1038	43	0.8837	0.9535	0.8837
3	1038	43	0.7674	0.8372	0.7674
4	1038	43	0.9070	0.9302	0.9070
5	1038	43	0.6512	0.6744	0.6512

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	3
Double_Nothing	0.500	0.333	0.400	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	0.667	1.000	0.800	4
Double_Wiggle	0.600	1.000	0.750	3
Double_cmere	1.000	0.500	0.667	4
Drum_Roll	1.000	0.333	0.500	3
macro avg	0.795	0.738	0.715	24
weighted avg	0.799	0.750	0.724	24

Batch test per class (Anghad)

Class	Correct	Total	Accuracy
Double_Lower	2	5	0.400
Double_Nothing	0	5	0.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	6	6	1.000
Drum_Roll	4	5	0.800
Overall	27	36	0.750

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9165 ± 0.0380
Hold-out test acc / loss	0.8750 · 0.6679
Batch-test acc	0.8889 (32/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9545	0.9545	0.9545
2	1038	43	0.9302	0.9302	0.9302
3	1038	43	0.8605	0.8837	0.8605
4	1038	43	0.9535	0.9535	0.9535
5	1038	43	0.8837	0.9767	0.8837

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	3
Double_Nothing	0.000	0.000	0.000	3
Double_Pistol_Recoil	0.667	1.000	0.800	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.750	1.000	0.857	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.774	0.857	0.808	24
weighted avg	0.788	0.875	0.824	24

Batch test per class (Anghad)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	1	5	0.200
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	6	6	1.000
Drum_Roll	5	5	1.000
Overall	32	36	0.889

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9490 ± 0.0400
Hold-out test acc / loss	0.9167 · 0.1766
Batch-test acc	0.8056 (29/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9545	0.9545	0.9545
2	1038	43	0.9767	1.0000	0.9767
3	1038	43	0.8837	0.8837	0.8837
4	1038	43	1.0000	1.0000	1.0000
5	1038	43	0.9302	0.9535	0.9302

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	3
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.750	1.000	0.857	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.936	0.905	0.892	24
weighted avg	0.935	0.917	0.901	24

Batch test per class (Anghad)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	1	5	0.200
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	6	6	1.000
Drum_Roll	2	5	0.400
Overall	29	36	0.806

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.8841 ± 0.0392
Hold-out test acc / loss	0.9167 · 0.5044
Batch-test acc	0.8333 (30/36)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9091	0.9318	0.9091
2	1038	43	0.9302	0.9302	0.9302
3	1038	43	0.8372	0.8837	0.8372
4	1038	43	0.8372	0.9535	0.8372
5	1038	43	0.9070	0.9070	0.9070

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	3
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.750	1.000	0.857	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.936	0.905	0.892	24
weighted avg	0.935	0.917	0.901	24

Batch test per class (Anghad)

Class	Correct	Total	Accuracy
Double_Lower	4	5	0.800
Double_Nothing	0	5	0.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	6	6	1.000
Drum_Roll	5	5	1.000
Overall	30	36	0.833